

Assignment 3: IEEE 754 Floating-Point Representation and Arithmetic

Auqib Hamid Lone

Department of Computer Science & Engineering, GCET Ganderbal

August 17, 2026

Deadline: two weeks from the date of issue — there is no automatic late penalty; if you need more time, ask before the deadline and an extension will be granted (see the course policy in the syllabus).

Answer every question. Show all working for Numerical and Design questions — a bare final answer without the derivation earns partial credit only.

3.1 Conceptual

Q1. The true exponent of a floating-point number can be negative, and IEEE 754 stores it *biased* ($E_{\text{stored}} = E_{\text{true}} + 127$ for single precision) rather than in two’s complement. Explain why the bias is used: what property of ordering the biased representation gives the hardware, and why that matters for comparison instructions. Your answer must state what a stored exponent of all zeros and all ones is *reserved* for.

Q2. A student writes a loop that accumulates $x = x + 0.1$ ten times and then tests $x = 1.0$. Explain why this test can fail even though $0.1 \times 10 = 1.0$ in exact arithmetic, and state the correct way to compare two floating-point numbers for “equality.”

3.2 Numerical

Q3. Encode the value 7.75 in IEEE 754 single precision (32 bits), showing: the binary expansion, the normalized scientific form $1.xxx_2 \times 2^E$, the sign bit, the biased exponent, and the 23-bit mantissa field. Write the final answer as an 8-hex-digit value (e.g. 0x412C0000).

Q4. Decode the 32-bit pattern 0x3F400000 into its decimal value, showing each field: sign, exponent, mantissa. Check your work by confirming the value against its binary scientific form. (Recall the lecture’s anchor: 0x3F800000 = 1.0.)

3.3 Design

Q5. Perform the single-precision addition $10.5 + 0.25$ qualitatively following the four-step add algorithm (align, add, normalize, round). For each step state precisely what happens to the mantissas and exponents. Then repeat for $1\,000\,000.0 + 0.25$, and explain, in terms of the alignment step, why the 0.25 may be lost entirely — the same effect the lecture demonstrated with 0.1.

Q6. Derive the single-precision product of 1.5×2.0 using the multiplication rule (multiply mantissas, add exponents, subtract the bias once, normalize, round). State the exact product’s

sign, exponent, and mantissa fields, and show that the result re-encodes to exactly 3.0. Explain why multiplication needs no alignment step.